KAIZEN MATHS

Lesson starter

Sampling Methods and Bias

Evaluate & Improve

Name

S1

Critique each sampling plan, then suggest one improvement.

Critique the sampling plan. Give two limitations and one improvement.A survey asks, 'Do you agree that the excellent new park should stay open late?'

S2

Design a suitable sampling plan and justify it.

Design a sampling plan and explain why it is suitable.A study needs reliable information about views on homework from all students in a secondary school. The population can be split by year group.

S3

Explain the bias and give a precise improvement.A transport survey is carried out only at a train station at 8:00 am.

S4

Critique each sampling plan, then suggest one improvement.

Critique the sampling plan. Give two limitations and one improvement.An interviewer explains one option enthusiastically but gives the other option very briefly.

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Show your working

KAIZEN MATHS

Exit ticket

Sampling Methods and Bias

Evaluate & Improve

Name _____

EA1

Design a suitable sampling plan and justify it.

Design a sampling plan and explain why it is suitable.A study needs reliable information about access to public transport from all households in a town. The population can be split by postcode area.

EA2

Explain the bias and give a precise improvement.A questionnaire asks: "How much do you hate the new homework policy?"

KAIZEN MATHS

Exit ticket

Sampling Methods and Bias

Evaluate & Improve

Name _____

EB1

Critique each sampling plan, then suggest one improvement.

Critique the sampling plan. Give two limitations and one improvement.A school posts an online survey about homework and only students who feel strongly decide to respond.

KAIZEN MATHS

Exit ticket

Sampling Methods and Bias

Evaluate & Improve

Name _____

EA1

Design a suitable sampling plan and justify it.

Design a sampling plan and explain why it is suitable.A study needs reliable information about access to public transport from all households in a town. The population can be split by postcode area.

EA2

Explain the bias and give a precise improvement.A questionnaire asks: "How much do you hate the new homework policy?"

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KAIZEN MATHS

Exit ticket

Sampling Methods and Bias

Evaluate & Improve

Name _____

EB1

Critique each sampling plan, then suggest one improvement.

Critique the sampling plan. Give two limitations and one improvement.A school posts an online survey about homework and only students who feel strongly decide to respond.

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Show your working

Teacher answers & working

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S1

Critique each sampling plan, then suggest one improvement.

Critique the sampling plan. Give two limitations and one improvement.A survey asks, 'Do you agree that the excellent new park should stay open late?'

Answer:

Limitations should include question wording bias. Improvement: Use neutral wording, such as 'Should the new park stay open late?'

Working:

A critique should refer to the target population, who is included, and who may be missed.

Limitation 1: The word 'excellent' pushes respondents towards a positive answer.

Limitation 2: the results may not generalise to the whole target population.

Improvement: Use neutral wording, such as 'Should the new park stay open late?'

The improved method is stronger because it reduces bias and gives a wider part of the population a chance to be selected.

S2

Design a suitable sampling plan and justify it.

Design a sampling plan and explain why it is suitable.A study needs reliable information about views on homework from all students in a secondary school. The population can be split by year group.

Answer:

Use stratified random sampling by year group, then randomly select within each stratum.

Working:

The important groups are the year group groups, so they should be represented.

Split the population into strata using year group.

Decide the total sample size, then calculate each group sample in proportion to its population size.

Use random selection within each group to choose the required number of people.

This is suitable because it combines proportional representation with random selection.

Teacher answers & working

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S3

Explain the bias and give a precise improvement.A transport survey is carried out only at a train station at 8:00 am.

Answer:

Selection Bias. Use several locations and times, then select people randomly from the wider passenger population.

Working:

Name the type of bias as precisely as possible.

Why it is biased: People at a train station at 8:00 am are likely to be commuters, so other travellers are under-represented.

Precise improvement: Use several locations and times, then select people randomly from the wider passenger population.

The improvement is better if it changes the method or wording directly, rather than only saying 'ask more people'.

S4

Critique each sampling plan, then suggest one improvement.

Critique the sampling plan. Give two limitations and one improvement.An interviewer explains one option enthusiastically but gives the other option very briefly.

Answer:

Limitations should include interviewer bias. Improvement: Use a standard script and train interviewers to present options neutrally.

Working:

A critique should refer to the target population, who is included, and who may be missed.

Limitation 1: The interviewer may influence responses through tone or explanation.

Limitation 2: the results may not generalise to the whole target population.

Improvement: Use a standard script and train interviewers to present options neutrally.

The improved method is stronger because it reduces bias and gives a wider part of the population a chance to be selected.

Teacher answers & working

Sampling Methods and Bias · Evaluate & Improve

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EA1

Design a suitable sampling plan and justify it.

Design a sampling plan and explain why it is suitable.A study needs reliable information about access to public transport from all households in a town. The population can be split by postcode area.

Answer:

Use stratified random sampling by postcode area, then randomly select within each stratum.

Working:

The important groups are the postcode area groups, so they should be represented.

Split the population into strata using postcode area.

Decide the total sample size, then calculate each group sample in proportion to its population size.

Use random selection within each group to choose the required number of people.

This is suitable because it combines proportional representation with random selection.

EA2

Explain the bias and give a precise improvement.A questionnaire asks: "How much do you hate the new homework policy?"

Answer:

Questionnaire Wording Bias. Rewrite it as: "What is your opinion of the new homework policy?"

Working:

Name the type of bias as precisely as possible.

Why it is biased: The wording assumes the respondent dislikes the policy.

Precise improvement: Rewrite it as: "What is your opinion of the new homework policy?"

The improvement is better if it changes the method or wording directly, rather than only saying 'ask more people'.

Teacher answers & working

Sampling Methods and Bias · Evaluate & Improve

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EB1

Critique each sampling plan, then suggest one improvement.

Critique the sampling plan. Give two limitations and one improvement.A school posts an online survey about homework and only students who feel strongly decide to respond.

Answer:

Limitations should include volunteer response bias. Improvement: Select students randomly and follow up selected students to improve response rate.

Working:

A critique should refer to the target population, who is included, and who may be missed.

Limitation 1: People with strong opinions are more likely to volunteer, so the results may not represent all students.

Limitation 2: the results may not generalise to the whole target population.

Improvement: Select students randomly and follow up selected students to improve response rate.

The improved method is stronger because it reduces bias and gives a wider part of the population a chance to be selected.